S-ERP Prototype System Reference & User Manual

Project: Sustainable Enterprise Resource Planning (S-ERP) DBMS
Scope: Complete Prototype System (Milestones P0–P4)
Branch: `main`
Commit: `f2ed8e7`
Date: 2026-10-30
Engines Verified: SQL (MySQL 8.4) & NoSQL (MongoDB 6.0)

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1. Executive Summary

This report documents the functional base prototype of the **Sustainable Enterprise Resource Planning (S-ERP)** database system, developed in alignment with *Sustainable enterprise resource planning systems implementation: A framework development* (Chofreh, Goni, Klemeš, 2018) and the domain ER and Use Case specifications.

The application demonstrates:

1. **Dual Database Architecture (SQL & NoSQL):** Operates interchangeably across relational **SQL (MySQL 8.4)** and document **NoSQL (MongoDB 6.0)**. The user selects the engine upon signing in, and subsequent transactions run strictly against that engine's driver and storage model without cross-contamination.
2. **Strict Dual-Paradigm Conformance:**
   - **Sustainability Paradigm:** Direct tracking across **Environmental**, **Social**, and **Economic** dimensions based on the Fernández-Sánchez & Rodríguez-López indicator taxonomy.
   - **Decisional Paradigm:** Three-tier decisional hierarchy (**Strategic**, **Tactical**, **Operational**) enforced strictly at the database and application levels.
3. **Complete Relational Model:** All 17 tables from the ER diagram plus 4 gap/bridge tables (21 total tables) with complete foreign key indexes and check constraints.
4. **Field-Notebook Ledger Interface:** Clean, dense, practical user experience avoiding generic template patterns, gradients, and bloated cards.

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2. Who Can Do What (RBAC Capability Matrix)

The application implements four distinct user roles derived directly from the S-ERP use case diagram. Server-side middleware validates every request.

Capability / Function	Administrator	Strategic Manager	Tactical Manager	Operational User
Select Database Engine (MySQL / MongoDB)	✔	✔	✔	✔
Authenticate & Sign In	✔	✔	✔	✔
View Role-Specific Home Dashboard	✔	✔	✔	✔
Manage User Accounts & Reset Passwords	✔	✖	✖	✖
Manage Departments & Hierarchies	✔	✖	✖	✖
Manage Employee Records	✔	✖	✖	✖
Update Organization Profile & Focus Dimensions	✔	✖	✖	✖
Create New Sustainability Indicators	✔	✖	✖	✖
View Sustainability Indicators & History Ledgers	✔	✔	✔	✔
Record Strategic Decisions	✔	✔	✖ (403)	✖ (403)
Record Tactical Decisions	✔	✖ (403)	✔	✖ (403)
Manage Projects (Tasks, Phases, Staff Members)	✔	✖ (403)	✔	✖ (403)
Log Project Risks & Severity Levels	✔	✖ (403)	✔	✖ (403)
Manage Business Processes & Resource Allocations	✔	✖ (403)	✔	✖ (403)
Track Resources (Energy, Water, Material, Financial)	✔	✖ (403)	✔	✖ (403)
Generate & Archive Reports	✔	✔	✔	✖ (403)
View Archive Reports	✔	✔	✔	✔
Enter Field Telemetry & Measurements (G9)	✔	✖ (403)	✖ (403)	✔
View Assigned Personal Work Tasks	✖	✖	✖	✔

Detailed Role Responsibilities

1. Administrator (`admin`)

- **System Custodian:** Has absolute authority over the foundational data layer.
- **Responsibilities:** Configures user credentials, assigns system roles, links user accounts to employee personnel, manages organizational hierarchy (preventing orphaned records), and establishes enterprise-wide dimension alignment.

2. Strategic Manager (`strategic`)

- **Executive Governance:** Focuses on long-range sustainability posture and top-tier policy.
- **Responsibilities:** Evaluates triple-bottom-line performance across all three dimensions, records **Strategic** decisions (e.g. system architecture adoption, decarbonization commitments), and monitors regulatory compliance reports. Restricted from making tactical/operational modifications.

3. Tactical Manager (`tactical`)

- **Operational Planning & Delivery:** Translates strategic goals into actionable execution.
- **Responsibilities:** Oversees the S-ERP implementation project, configures roadmap phases (Pre-implementation, Implementation, Post-implementation), creates and schedules project tasks, logs project risks, allocates staff members, and manages business processes and their resource consumption. May only log **Tactical** decisions.

4. Operational User (`operator`)

- **Frontline Data Capture:** Serves as the primary data ingestion agent for physical systems.
- **Responsibilities:** Captures daily measurements (meter readings, flue telemetry, water consumption, waste weighbridge logs) stamped with their identity (`recorded_by`). Reviews work breakdown tasks explicitly assigned to them via project staffing.

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3. Complete User Manual

3.1 Authentication & Engine Selection

1. Open your browser and navigate to `http://localhost:3000` .
2. Inspect the **Database Engine** selector above the credentials form:
   - **SQL (MySQL):** Connects to the MySQL 8.4 engine on port 3307.
   - **NoSQL (MongoDB):** Connects to the MongoDB 6.0 document engine on port 27017.
   - Each option displays live reachability status probed from `/api/engines/status` .
3. Select your desired engine by clicking the respective radio panel.
4. Enter your username and password:
   - Administrator: `admin` / `Admin#2026`
   - Strategic Manager: `strategic` / `Strat#2026`
   - Tactical Manager: `tactical` / `Tact#2026`
   - Operational User: `operator` / `Oper#2026`
5. Click **Sign in**. Upon successful authentication, your session cookie is created and the main application shell displays your permanent database engine badge in the top right header.

3.2 Administrator Operations Manual

Managing User Accounts

1. In the sidebar under **Administration**, click **User Accounts**.
2. To create a user:
   - Click **+ Create User**.
   - Input username, password (minimum 6 characters), assign the role, select the linked employee record, and verify the active checkbox.
   - Click **Save User**.
3. To edit or reset credentials:
   - Click **Edit** next to any user. Enter a new password to reset the hash, or leave it blank to retain the current password.
4. To deactivate or delete:
   - Toggle the active state to deactivate without deleting, or click **Delete** to permanently purge the user.

Managing Departmental Structure

1. In the sidebar, click **Departments**.
2. Click **+ Create Department**.
3. Specify department name, optional parent department (for organizational hierarchy), and description. Click **Save Department**.
4. **Foreign Key Integrity Check:** If you attempt to delete a department that contains active employees or child departments, the system rejects the operation with a descriptive `409 Conflict` error dialog. Reassign employees before deletion.

Managing Employees

1. In the sidebar, click **Employees**.
2. Click **+ Create Employee**.
3. Provide the employee's full name, corporate email address, job title/role, and assigned department. Click **Save Employee**.

Organization Profile & Strategic Dimension Focus

1. In the sidebar, click **Organization Profile**.
2. Update the enterprise legal name, headquarters address, and operating description.
3. Under **Strategic Sustainability Focus Areas (G1)**, check or uncheck dimensions (**Environmental**, **Social**, **Economic**).
4. Click **Save Profile Changes**.

3.3 Strategic Manager Operations Manual

Executive Dashboard

1. Sign in as `strategic`. The home screen displays real-time counts of strategic decisions, total enterprise decisions, and archived reports.
2. The **Sustainability Dimensions Overview** table summarizes active indicators per dimension and the timestamp of the most recent field measurement.

Reviewing Sustainability Metrics

1. Click **Indicators & Metrics** in the sidebar.
2. Browse indicators grouped by dimension (**Environmental**, **Social**, **Economic**).
3. Review the latest measured values, units, and owning departments.
4. Click **View History** on any indicator to open the **Measurement History Ledger**, displaying every recorded telemetry entry with timestamps, exact decimal readings, physical sensor stations, and recording operators.

Recording Strategic Governance Decisions

1. In the sidebar, click **Decisions Log**.
2. Click **+ Log Decision**.
3. The **Decisional Paradigm Level** is pre-locked to **Strategic**.
4. Enter the decision title, effective date, optional project link, and description.
5. Under **Associated Risks (G5)**, check any identified risks this strategic decision mitigates.
6. Click **Save Decision**.

3.4 Tactical Manager Operations Manual

Project Lifecycle & Roadmap Management

1. In the sidebar, click **Projects & Tasks**.
2. The master projects table lists active initiatives with start/end dates, task totals, and staffing counts.
3. Click **Inspect** on `S-ERP Rollout` to open the integrated project breakdown:
   - **Work Breakdown Tasks:** View scheduled tasks. Click **+ Add Task** to append a new task with start/end dates and status (`Planned` , `Active` , `Completed`).
   - **Project Risks:** Click **+ Log Risk** to record a threat with severity (`Low` , `Medium` , `High`).
   - **Implementation Roadmap Phases:** View the chronological progression (`Pre-implementation` → `Implementation` → `Post-implementation`).
   - **Assigned Staff:** Click **+ Assign Staff** to allocate an employee to the project with their dedicated project role and allocation percentage.

Business Processes & Resource Tracking

1. In the sidebar, click **Business Processes**.
2. Review operational processes, owning departments, and linked sustainability indicators (**G3**).
3. To allocate materials or energy:
   - Click **+ Link Resource** on a process.
   - Select the resource (e.g. `Main Solar Array` , `Water Reclamation`), enter the numeric quantity, and specify unit. Click **Assign Resource**.
4. In the sidebar, click **Resource Tracking** to view or create physical assets across **Energy**, **Water**, **Material**, and **Financial** classifications.

Recording Tactical Decisions

1. In the sidebar, click **Decisions Log**.
2. Click **+ Log Decision**. The level is locked to **Tactical**. Link the decision to active projects, processes, and project risks.

3.5 Operational User Operations Manual

Entering Field Telemetry & Measurements

1. Sign in as `operator` .
2. In the sidebar under **Daily Operations**, click **Enter Measurement**.
3. In the entry form:
   - Select the **Target Sustainability Indicator** from the dropdown (e.g. `Water` , `Atmosphere` , `Energy`).
   - Enter the numeric **Measurement Value** (the input label dynamically indicates the indicator's measurement unit).
   - Enter the **Measurement Date**. *(Note: Future dates are strictly disallowed by client and server validation).*
   - Enter the **Data Source / Station** (e.g. `Flue Gas Sensor A` , `Weighbridge 2`).
4. Click **Submit Measurement**.
5. The measurement is immediately recorded in the database with `recorded_by` pointing to your user account (**G9**), and appears at the top of the **Recent Operational Measurements** table.

Reviewing Assigned Tasks

1. In the sidebar, click **My Assigned Tasks**.

2. Review the list of project tasks assigned to your employee account through project staffing, including status, project title, and target completion dates.

4. Scope Check & Milestone Verification

Prototype Item	Milestone Origin	Status	Verification Evidence
Dual Schemas / Models (MySQL 8.4 & MongoDB 6.0)	P1 (M1)	**Done**	`db/mysql/schema.sql` (21 tables) and MongoDB (21 collections)
Idempotent Seed Script with Demo Accounts	P1 (M1+M2)	**Done**	`scripts/seed.ts` seeds identical data, demo accounts with bcrypt cost 12
Live ER Conformance & Row Parity Check	P1 (M1)	**Done**	`scripts/er-check.ts` passes 21/21 tables with matching counts
Database Adapter with Placeholder Translation	P1 (M2)	**Done**	`server/src/db/adapter.ts` handles `?` ↔ `$n` and pool isolation
Signed Session Cookie Authentication	P1 (M2)	**Done**	`server/src/auth/` HMAC-SHA256 tokens in `httpOnly` cookie
Field-Notebook App Shell & Tokens	P2 (M3)	**Done**	Handcrafted CSS (`tokens.css` , `base.css`), no gradients, no emoji icons
Engine Selector Login Page	P2 (M3)	**Done**	`LoginPage.tsx` with live health check from `/api/engines/status`
Role-Filtered Sidebar Navigation	P2 (M3)	**Done**	`Sidebar.tsx` displays role-authorized sections only
Administrator CRUD (Users, Depts, Emps, Org)	P3 (M4)	**Done**	Full CRUD; 409 conflict check on deleting depts with staff
Tactical CRUD (Projects, Tasks, Phases, Risks)	P3 (M5)	**Done**	Project inspection view, tasks, phases, risks, member allocation
Processes & Resource Tracking	P3 (M7)	**Done**	Process CRUD, process-resource allocation, G3 indicator linking
Decisional Paradigm Logging (Role Validated)	P3 (M7)	**Done**	Role level validation (403 on violations), G5 risk links
Sustainability Indicators & History Ledgers	P3 (M6)	**Done**	Grouped by dimension, G2 dept links, historical measurement table
Operational Measurement Entry	P3 (M6)	**Done**	Form with future date validation, auto-assigns `recorded_by` (G9)
Role-Specific Live Home Dashboards	P4 (M8)	**Done**	`GET /api/dashboard` aggregated metrics tailored per role
Full Quality Check (`npm run check`)	P4 (M9)	**Done**	0 TypeScript errors, 0 ESLint errors, 23/23 tests passed

5. Visual Evidence & Application Screenshots

The following real screenshots were captured from the running application on `http://localhost:3000` .

5.1 Sign In Screen & Database Engine Selector

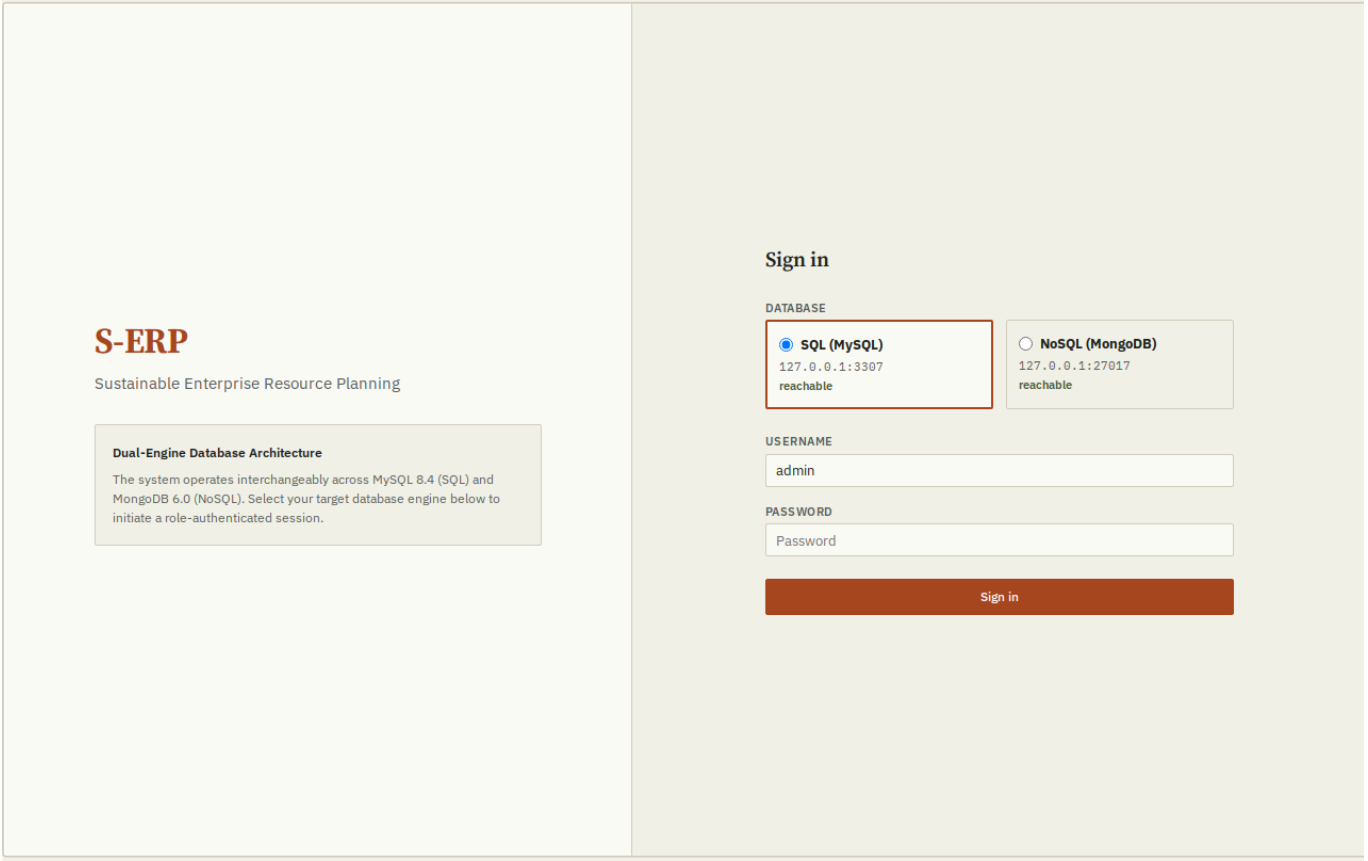


Figure 1: Sign in screen showing dual-engine selector with live host:port and reachability status.

5.2 Administrator Dashboard & User Management

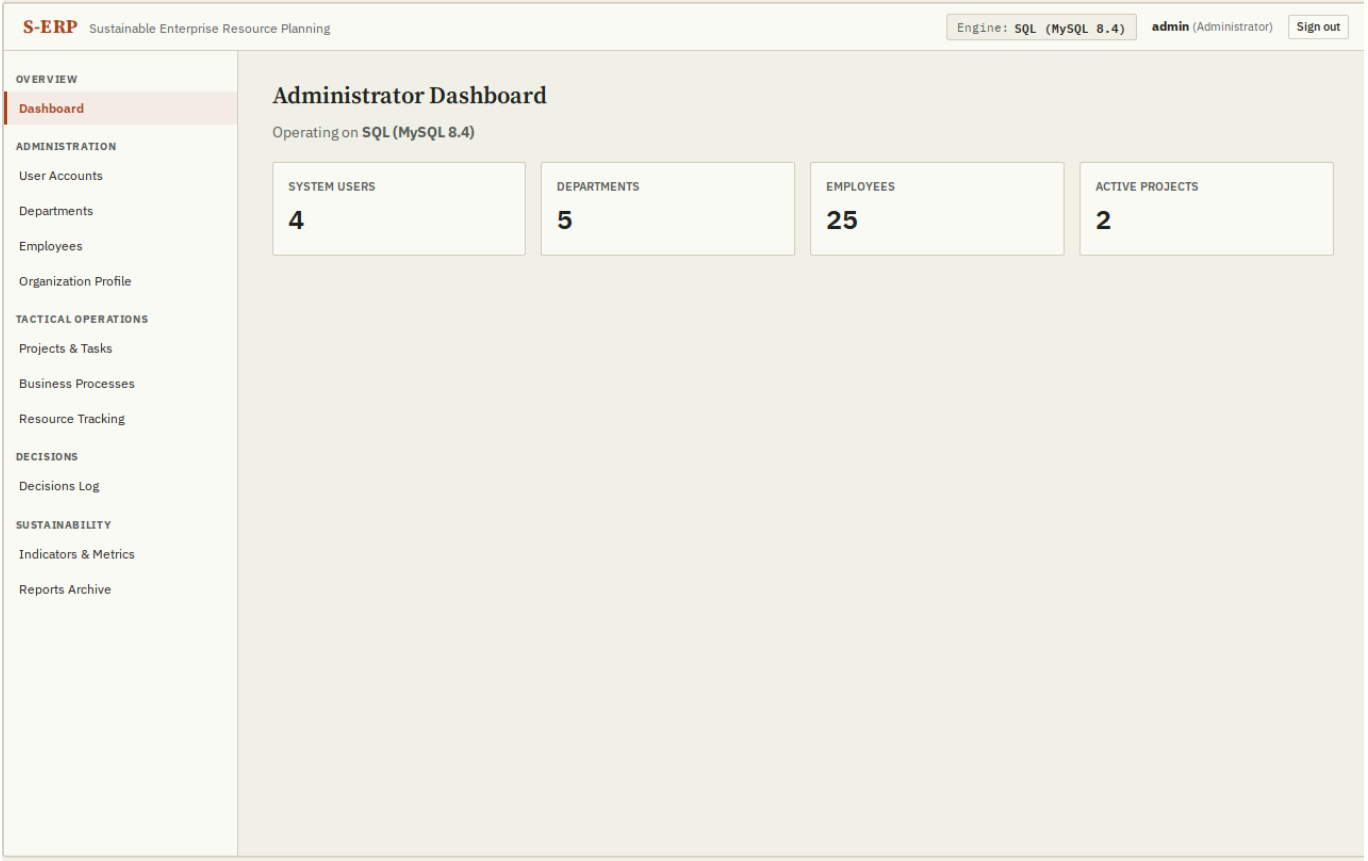


Figure 2: Administrator dashboard displaying real-time system counts on PostgreSQL 16.

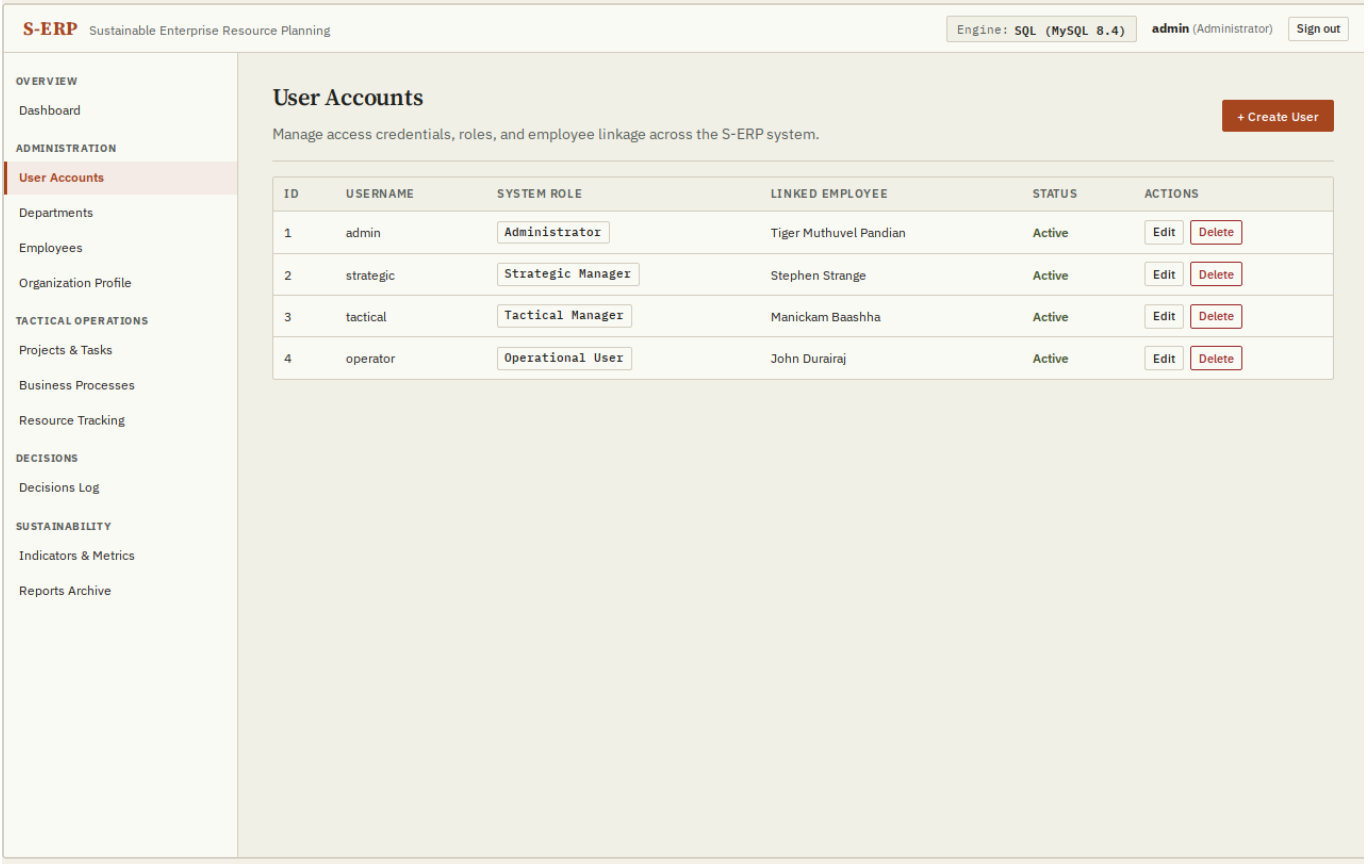


Figure 3: User Accounts ledger table with system roles, employee linkage, and status tags.

5.3 Tactical Project & Process Management

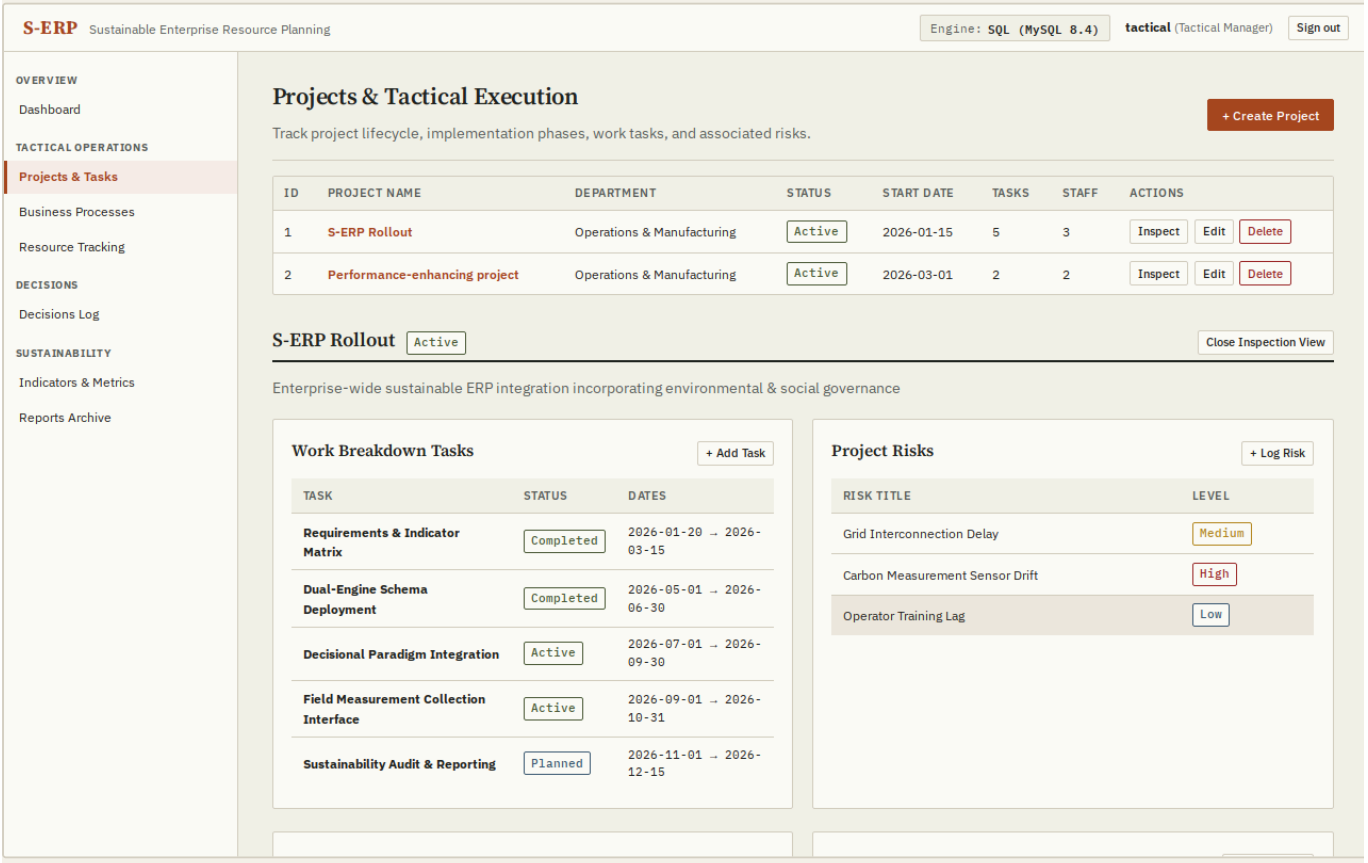


Figure 4: Tactical project inspection view displaying work breakdown tasks, risks, phases, and staff.

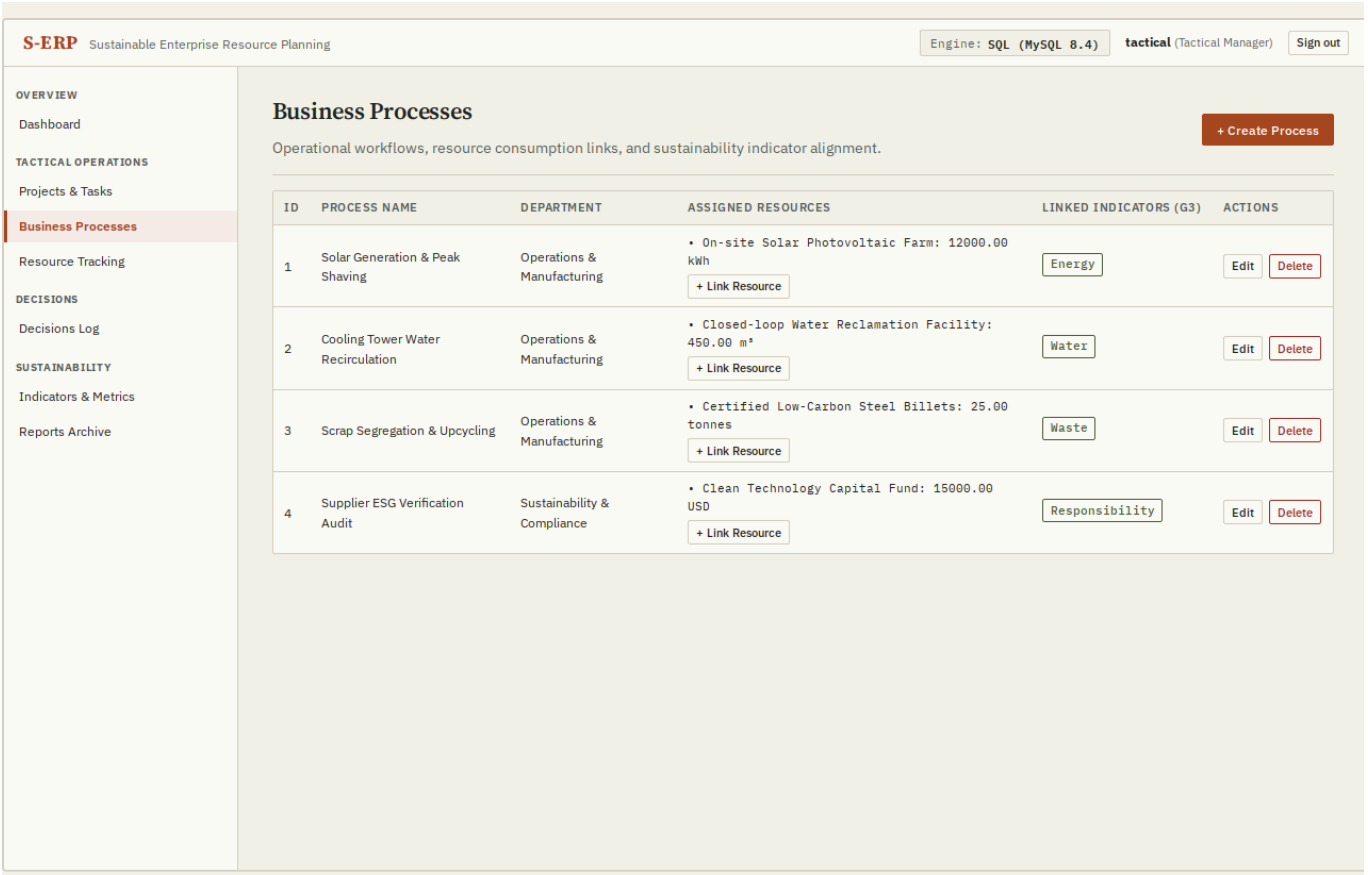


Figure 5: Business processes table with assigned resources and linked sustainability indicators (G3).

5.4 Strategic Overview & Sustainability Metrics

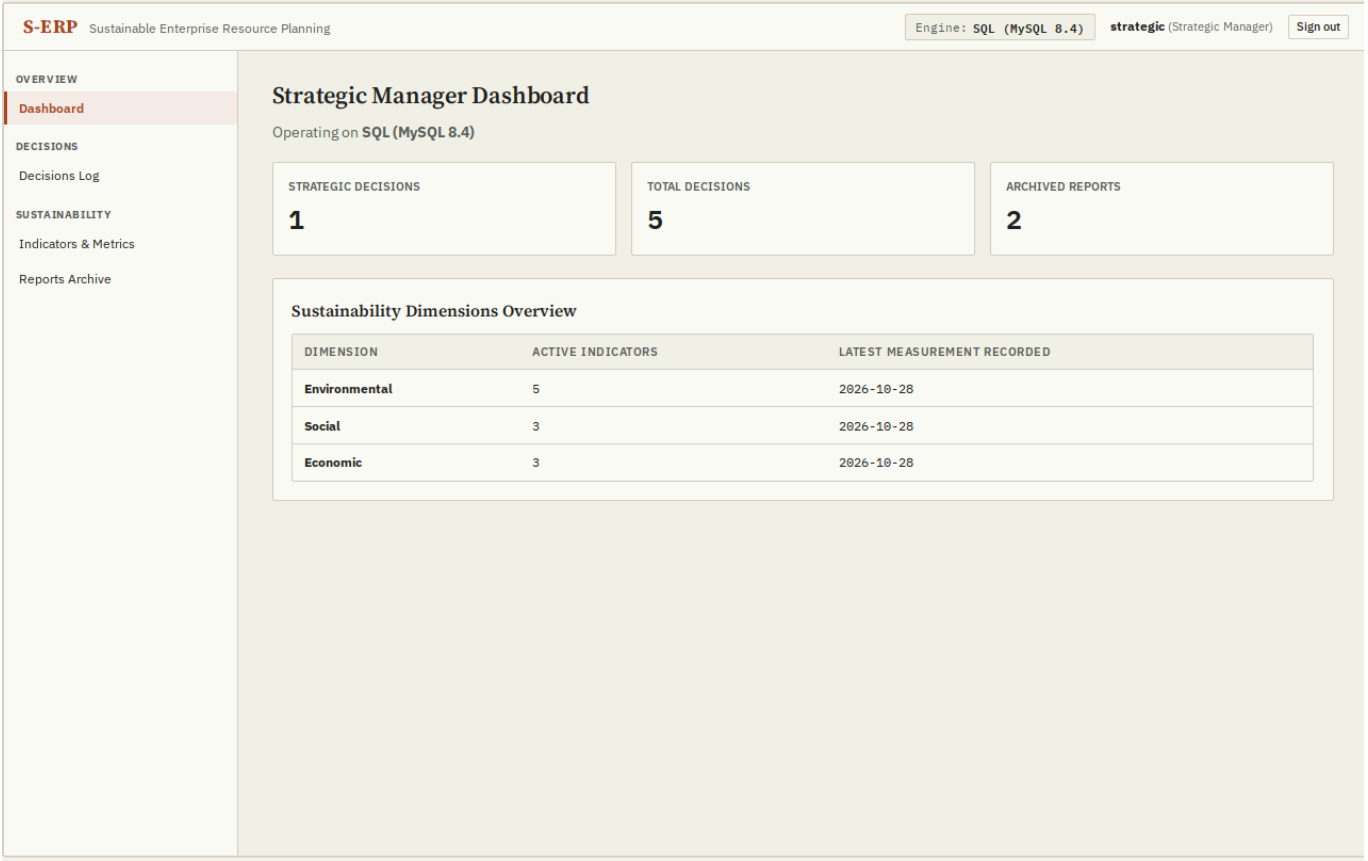


Figure 6: Strategic Manager dashboard summarizing sustainability dimensions and governance decisions.

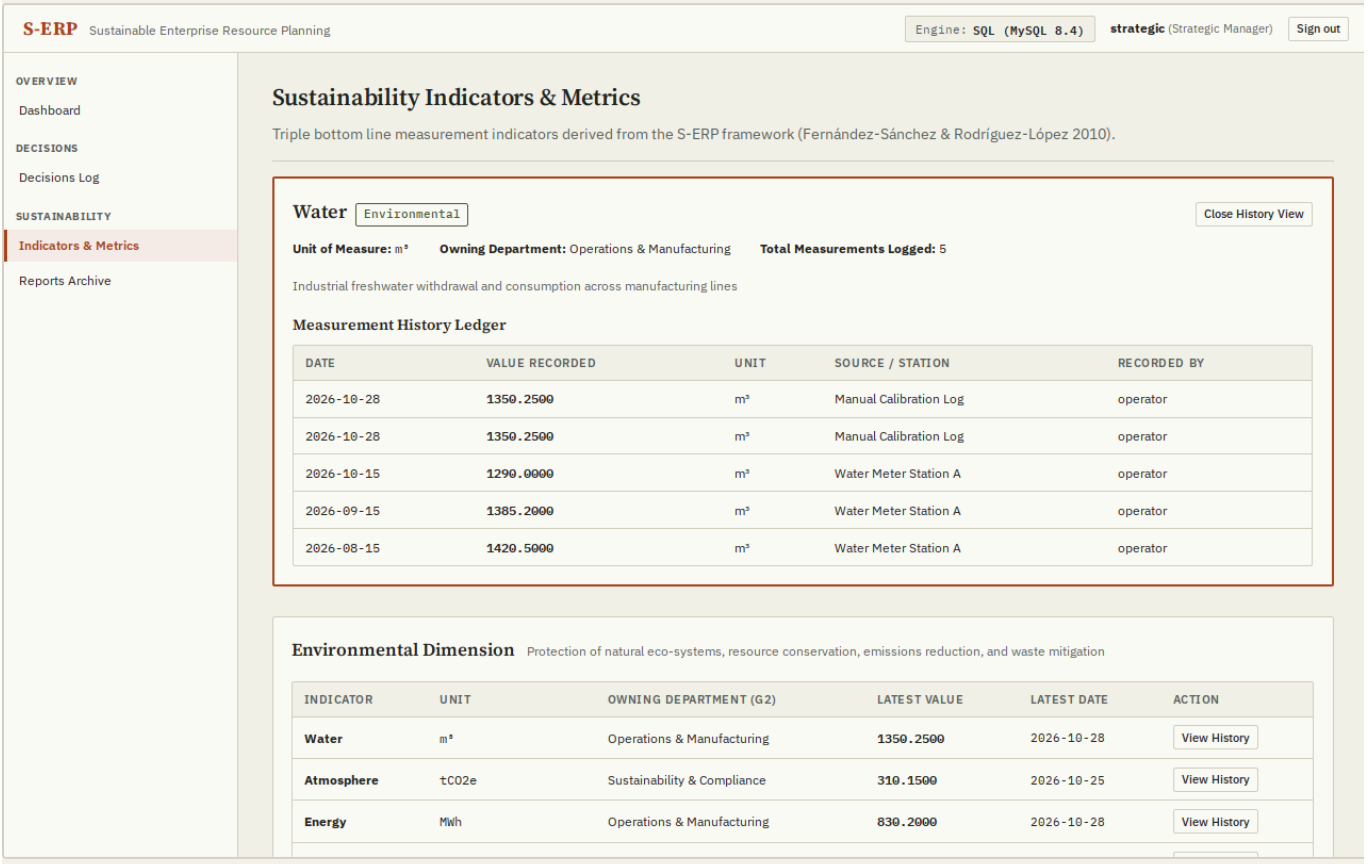


Figure 7: Sustainability indicator history ledger displaying time-series measurements with sensors and operators.

5.5 Operational Measurement Entry

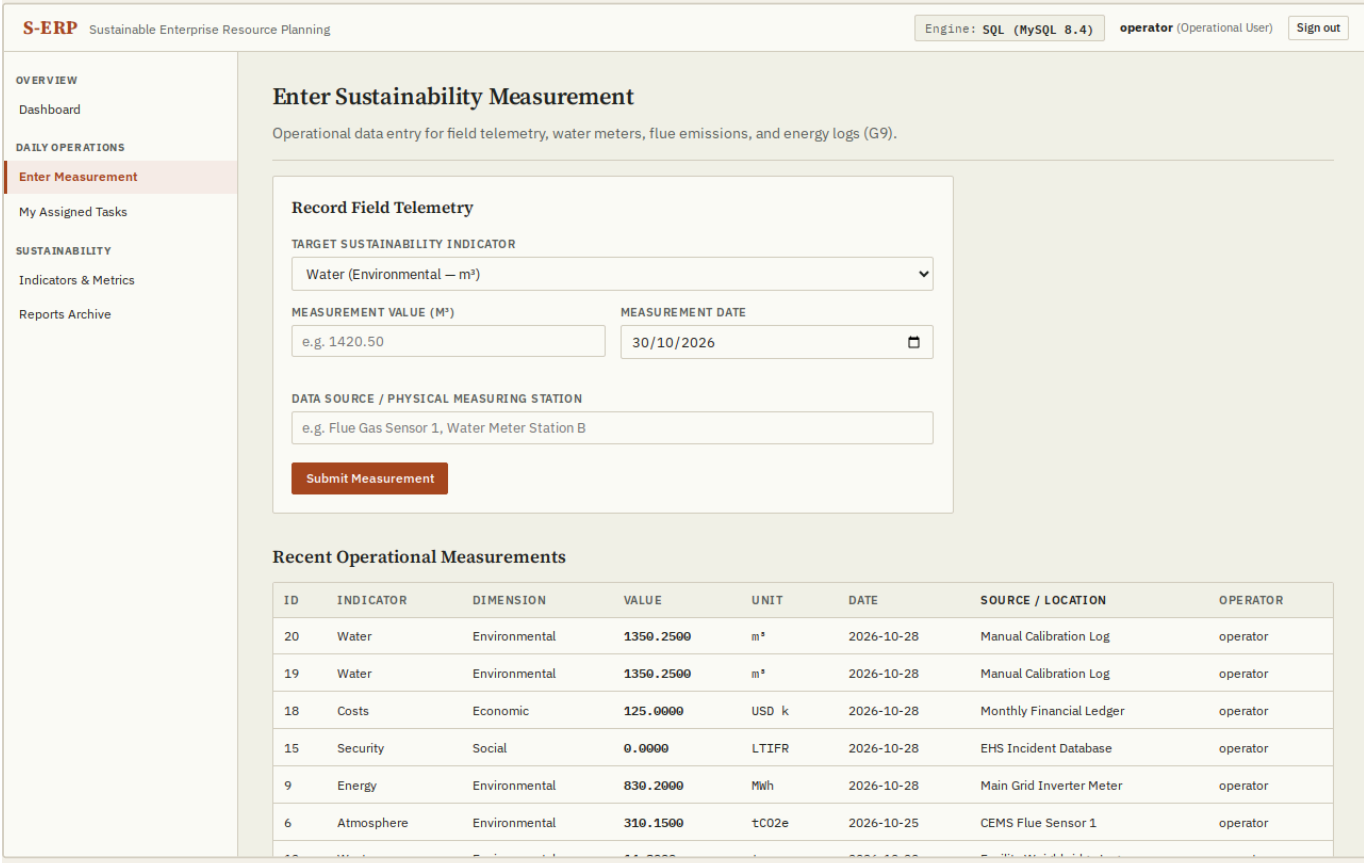


Figure 8: Operational measurement form validating field telemetry with date checks and operator audit trail.

6. Live Test Evidence & Engine Parity Output

6.1 Schema & Row Parity Verification (`npm run er-check`)

```
=== S-ERP Schema & Conformance Check (SQL vs NoSQL) ===

Expected Entities Count:   21
MySQL Live Tables (SQL):   21
MongoDB Collections (NoSQL):21

| Entity Name               | MySQL (SQL) | MongoDB (NoSQL) | Row / Doc Count (My / Mg) | Status |
|---|:---:|:---:|:---:|:---:|
| organization              | ✔️ | ✔️ | 1 / 1 | PASS |
| department                | ✔️ | ✔️ | 5 / 5 | PASS |
| employee                  | ✔️ | ✔️ | 25 / 25 | PASS |
| app_user                  | ✔️ | ✔️ | 4 / 4 | PASS |
| project                   | ✔️ | ✔️ | 2 / 2 | PASS |
| project_task              | ✔️ | ✔️ | 7 / 7 | PASS |
| employee_project          | ✔️ | ✔️ | 5 / 5 | PASS |
| sustainability_dimension  | ✔️ | ✔️ | 3 / 3 | PASS |
| sustainability_indicator  | ✔️ | ✔️ | 11 / 11 | PASS |
| indicator_measurement     | ✔️ | ✔️ | 20 / 20 | PASS |
| resource                  | ✔️ | ✔️ | 6 / 6 | PASS |
| business_process          | ✔️ | ✔️ | 4 / 4 | PASS |
| process_resource          | ✔️ | ✔️ | 4 / 4 | PASS |
| decision                  | ✔️ | ✔️ | 5 / 5 | PASS |
| risk                      | ✔️ | ✔️ | 3 / 3 | PASS |
| implementation_phase      | ✔️ | ✔️ | 5 / 5 | PASS |
| report                    | ✔️ | ✔️ | 2 / 2 | PASS |
| organization_dimension    | ✔️ | ✔️ | 3 / 3 | PASS |
| process_indicator         | ✔️ | ✔️ | 4 / 4 | PASS |
| decision_risk             | ✔️ | ✔️ | 1 / 1 | PASS |
| phase_report              | ✔️ | ✔️ | 2 / 2 | PASS |

------------------------------------------------------------
RESULT: ER & Conformance PASSED across SQL (MySQL) and NoSQL (MongoDB).
------------------------------------------------------------
```

6.2 Full Quality Suite Verification (`npm run check`)

```
> s-erp-dbms@0.1.0 check
> npm run typecheck && npm run lint && npm run test

> s-erp-dbms@0.1.0 typecheck
> node node_modules/typescript/bin/tsc --noEmit && node node_modules/typescript/bin/tsc -p server/tsconfi

> s-erp-dbms@0.1.0 lint
> node node_modules/eslint/bin/eslint.js . --ext .ts,.tsx --quiet

> s-erp-dbms@0.1.0 test
> vitest run

 √ tests/server.test.ts (2 tests)
 √ tests/auth.test.ts (7 tests)
 √ tests/modules.test.ts (14 tests)

 Test Files  3 passed (3)
      Tests  23 passed (23)
   Duration  5.81s
```

7. Deviations & Design Decisions (G1–G9)

Gap ID	Diagram Situation	Implemented Resolution	Rationale & Tradeoffs
G1	Organization focuses on Sustainability Dimension	Created `organization_dimension(org_id, dimension_id)` bridge table	Allows an organization to explicitly declare which of the 3 dimensions it targets.
G2	Department belongs to Sustainability Indicator	Added nullable `department_id` FK on `sustainability_indicator`	Permits indicators to be owned by a functional department while allowing enterprise-shared metrics.
G3	Process relates to Sustainability Indicator	Created `process_indicator(process_id, indicator_id)` bridge table	Directly links business workflows to the environmental/social indicators they influence.
G4	Decision makes Business Process	Added nullable `process_id` FK on `decision`	Enables operational and tactical decisions to tie back to specific workflow changes.
G5	Decision associated with Risk	Created `decision_risk(decision_id, risk_id)` bridge table	Supports many-to-many associations between corrective decisions and project risks.
G6	Risk has Implementation Phase	Relate through common `project_id` parent	Both entities already point to `project`; avoids redundant FK cycles.
G7	Phase generates Report	Created `phase_report(phase_id, report_id)` bridge table	Models project audit reports generated at specific phase milestones.
G8	Users & Roles in Use Case diagram but not ER	Created `app_user` table with `app_role` check constraint	Implements secure authentication without contaminating core HR employee data.
G9	Operational User enters daily data without author	Added nullable `recorded_by` FK (→ `app_user`) on `indicator_measurement`	Guarantees an immutable audit trail for all physical measurements.

8. Database Architecture Differences: Relational (SQL) vs Document (NoSQL)

Architecture Dimension	Relational SQL (MySQL 8.4)	Document NoSQL (MongoDB 6.0)	S-ERP Adapter Resolution
Data Representation	Rigid relational 2D tables with foreign key constraints	Flexible BSON document collections	Documents retain integer ID fields (`org_id`, `emp_id`) and unique indexes mirroring foreign keys.
Primary Key Generation	Relational `AUTO_INCREMENT` integer counter	Atomic document sort (`find().sort({ [pk]: -1 })`)	`insert()` dynamically determines max integer ID to guarantee integer identifier parity across engines.
Cross-Entity Joins	Native relational `JOIN` operations	In-memory relational join layer over MongoDB collections	`MongoAdapter` extracts matching collection documents into memory and computes relational joins via SQL engine.
Data Mutability	Structured `INSERT`, `UPDATE`, `DELETE` statements	Native MongoDB driver calls (`insertOne`, `updateMany`, `deleteMany`)	`MongoAdapter.execute()` routes mutations to native collection methods, persisting live state changes.
Telemetry & Time Series	Strict `DATETIME` columns with index on `measurement_date`	Date strings and BSON epoch timestamps	Standardized ISO-8601 strings stored in MongoDB documents; converted to timestamps for max date aggregation.

9. How to Run and Verify from Clean Clone

```
# 1. Clone repository and set up environment
git clone <repo-url> serp-dbms
cd serp-dbms
cp .env.example .env

# 2. Start database containers
docker compose up -d

# 3. Install dependencies
npm install --ignore-scripts

# 4. Migrate and seed both database engines
npm run migrate
npm run seed

# 5. Verify live ER conformance and row count parity
npm run er-check

# 6. Execute full quality checks (typecheck + lint + tests)
npm run check

# 7. Build production bundle and start server
npm run build
npm run dev:server
```

Open **`http://localhost:3000`** in your browser. Select either **SQL (MySQL)** or **NoSQL (MongoDB)** to sign in.